

No totally positive 3×4 matrix over $\mathbb{F}_{19}$

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Abstract

Call an element of $\mathbb{F}_{19}$ positive when it is a nonzero square. We prove that no 3×4 matrix over this field has all its entries and all its minors of order two positive. Consequently, no such matrix is totally positive. Positive diagonal scaling reduces each of the two lower rows to one of two triples; an explicit minor excludes each of the four ordered pairs. An exhaustive exact computation of all $9^6 = 531\,441$ normalized matrices corroborates the elementary proof. The conclusion concerns this single field and matrix size.

1 Statement and context

All arithmetic below is in $\mathbb{F}_{19} = \mathbb{Z}/19\mathbb{Z}$. Its nonzero squares are

$$Q = \{1, 4, 5, 6, 7, 9, 11, 16, 17\}.$$

For equal-sized index sets I and J , the notation $A[I, J]$ denotes the submatrix with rows and columns in increasing original index order. In particular, a minor of order two means $A_{ik}A_{j\ell} - A_{i\ell}A_{jk}$ with $i < j$ and $k < \ell$, including nonconsecutive indices. Following Ayyer and Prasad [1, Definitions 2.6 and 5.1], a matrix is *totally positive* if every square minor is a nonzero square. Zero is not positive.

Theorem 1. *There is no $A \in M_{3,4}(\mathbb{F}_{19})$ for which every entry and every minor of order two belongs to Q . In particular, there is no totally positive 3×4 matrix over $\mathbb{F}_{19}$.*

Ayyer and Prasad establish positive diagonal normalization [1, Proposition 5.2] and state the necessary condition $q \geq 4(m+n) - 9$ for total positivity with $m, n \geq 2$ over $\mathbb{F}_q$ when $q \equiv 3 \pmod{4}$ [1, Proposition 5.6]. For $(m, n) = (3, 4)$, this bound gives $q \geq 19$. Theorem 1 excludes equality in this particular instance, already using only the first two orders of minors. It does not determine the smallest field supporting total positivity of this size or settle their general minimal-field question [1, Question 5.7].

Adjacent finite-field matrix context is provided by the work of Zeng, Liu and Ratnavelu on nowhere-zero vectors for invertible matrices [2]. That reference is contextual; the proof here has no dependence on a result about nowhere-zero vectors. The literature comparison is confined to the cited work and a bounded search conducted on 7 September 2026; no bibliographic priority is asserted.

2 Proof

Proof of Theorem 1. Suppose such a matrix A exists. Nonzero squares are closed under multiplication and inversion. Set

$$D = \text{diag}(A_{11}^{-1}, A_{21}^{-1}, A_{31}^{-1}), \quad E = \text{diag}\left(1, \frac{A_{11}}{A_{12}}, \frac{A_{11}}{A_{13}}, \frac{A_{11}}{A_{14}}\right), \quad B = DAE.$$

All the diagonal factors lie in Q . For any equal-sized index sets I, J ,

$$\det B[I, J] = \left(\prod_{i \in I} D_{ii} \right) \left(\prod_{j \in J} E_{jj} \right) \det A[I, J].$$

Thus a minor belongs to Q before scaling if and only if it does after scaling. Since $B_{ij} = A_{11}A_{ij}/(A_{i1}A_{1j})$, we obtain

$$B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & a & b & c \\ 1 & d & e & f \end{pmatrix}, \quad (a, b, c, d, e, f) \in Q^6. \quad (1)$$

No row or column permutation is used.

Consider either lower row, written $(1, x, y, z)$. The minors involving the first row and the first column require

$$x, y, z \in S := \{t \in Q : t - 1 \in Q\} = \{5, 6, 7, 17\}.$$

The other three minors involving the first row require $y - x, z - x, z - y \in Q$. The complete difference table on S is

$y - x$	$y = 5$	$y = 6$	$y = 7$	$y = 17$
$x = 5$	0	1	2	12
$x = 6$	18	0	1	11
$x = 7$	17	18	0	10
$x = 17$	7	8	9	0

(2)

where the bold entries are exactly those in Q . All entries in this table, and the set Q , follow by arithmetic modulo 19; squaring $1, \dots, 9$ gives $1, 4, 9, 16, 6, 17, 11, 7, 5$.

The entries x, y, z must be distinct, because zero is not in Q . The first entry must have two distinct successors in (2) with positive differences. Only 6 and 17 do. If $x = 6$, the other two entries are 7 and 17, and $z - y \in Q$ forces their order to be 17, 7. If $x = 17$, the other two are 5 and 7, and their order must be 7, 5. Hence the only possible free triples are

$$U = (6, 17, 7), \quad V = (17, 7, 5).$$

Both triples meet all conditions involving the first row: the six differences for the row $(1, U)$ are $5, 16, 6, 11, 1, 9$, and those for $(1, V)$ are $16, 6, 4, 9, 7, 17$. These lists use the column-pair order $(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)$.

It remains to check the four ordered pairs of lower rows. In the following table, every minor uses rows 2, 3, and U, V designate the free triples in columns 2, 3, 4:

Row 2 triple	Row 3 triple	Columns	Determinant modulo 19
U	U	1, 2	$6 - 6 = 0$
V	V	1, 2	$17 - 17 = 0$
V	U	1, 2	$6 - 17 = 8$
U	V	2, 3	$6 \cdot 7 - 17 \cdot 17 = 0$

For the final entry, $6 \cdot 7 - 17 \cdot 17 = -247 = -13 \cdot 19$. Neither 0 nor 8 belongs to Q , so every case contradicts a required minor of order two. This proves the theorem; no minor of order three is needed. $\square$

3 Exact computation and reproducibility

A direct enumeration examined every ordered tuple in Q^6 , with no filtering by S and no symmetry quotient. For each normalized matrix (1), it tested the 18 minors of order two in increasing row-pair and then column-pair order, stopping at the first failure. All 531 441 tuples failed. The accompanying compressed CSV records each tuple and the indices and residue of its first failing minor. A separately implemented replay used explicit squares, base-nine tuple indexing and modular elimination to validate every record. A further implementation evaluated all 18 minors on every tuple, making 9 565 938 evaluations, and found no survivor. These finite checks corroborate the proof, which does not require execution of a program.

The supplementary `verify.py` provides a compact read-only check. It reconstructs Q , verifies the two row triples and checks all 18 minors for all 9^6 normalized matrices using integer arithmetic. With the optional CSV argument, it also verifies all recorded first failures and the exact tuple order. From the project directory, run

```
python3 manuscript/verify.py evidence/full_exclusion.csv.gz
```

Python 3 and its standard library suffice. No floating-point arithmetic, random sampling or external solver is involved. Source files, the complete certificate and reproduction instructions accompany the local manuscript.

Computational and AI assistance. OpenAI Codex assisted with proof discovery, exact computation, checking and manuscript preparation. A separate model context reconstructed the argument and implemented a further exact verifier. These checks are not external peer review or proof-assistant formalization. The mathematical argument is given in full above.

References

- [1] Arvind Ayyer and Shubhanshu Prasad, *Positive definite, positive semidefinite and totally positive matrices over finite fields*, arXiv:2608.17702v1 (2026). <https://arxiv.org/abs/2608.17702v1>.
- [2] Zijian Zeng, Houde Liu and Kuru Ratnavelu, *Closing the Permanent-Rank Boundary for the Alon–Jaeger–Tarsi Conjecture*, Preprints (2026). <https://doi.org/10.20944/preprints202608.1194.v1>.